Simulating bearings-only tracking for a small drone: an EKF and CKF comparison harness

Python. 400 Monte Carlo runs per data point. Code, tests and figures in the accompanying repository.

This is a simulator, not a derivation. It exists to answer questions of the form “does this filter work on this geometry, and how do I know” quickly enough that the answer arrives while the question is still interesting. The filters are textbook implementations; the work is in the harness around them.

1. What this is

A small drone with a camera measures the direction to another aircraft and nothing else. There is no range. The question is whether a Kalman filter can turn a sequence of angles into a track worth using, and whether a cubature filter is worth running instead of an extended Kalman filter for that job.

Both filters are standard. The extended Kalman filter linearises the measurement about the current estimate; the cubature filter samples it at a fixed set of points instead. Neither is derived here and neither is modified. The degree-5 cubature rule is McNamee and Stenger's, applied to filtering by Jia, Xin and Cheng. What this project contributes is the apparatus for putting a scenario through both and getting an answer that can be trusted.

The reason that apparatus is the interesting part is that most of the effort in a study like this is not the estimator. It is deciding whether a scenario is worth running, whether a difference between two runs is real or noise, and whether the thing being measured is what was intended. Those are the parts that were built.

2. Cost of one iteration

Everything that can change a number lives on one frozen dataclass with 62 fields: geometry, kinematics, sensor aperture, detection probability, noise models, estimator options, evaluation thresholds. Nothing that affects a result is a literal buried in a module. A variant is `replace(scenario, p0_pos=600.0)`, and because the object is frozen a scenario cannot be mutated halfway through a sweep.

Table 1. Measured wall-clock cost of putting one question through the harness, on a laptop.

step	*time*
screen a scenario for viability, 8 runs	0.4 s
one estimator, 50 runs, first look	1.9 s
one estimator, 400 runs, publishable	15 s
three estimators across three scenarios, 400 runs	2.3 min
full test suite	2.9 min

The screening step is the one that saved the most time, and it exists because of repeated failures. Every scenario written for this project was broken at least once, and in each case the breakage was visible in four cheap numbers that nobody computed until after a full sweep had been run and interpreted: detection rate, range span, bearing sweep, track loss. A forty-degree target manoeuvre that lost 100 per cent of tracks. A sixty-degree fixed camera that detected the target on zero scans. A three-actor engagement that lost 69.5 per cent of its tracks, whose survivor statistics were nearly reported as a result. Two scenarios that were byte-identical because an override matched the default. All of those now fail in 0.4 seconds instead of after an hour of interpretation.

3. What keeps the answers honest

Four things, none of them clever, all of them load-bearing.

The simulator and the filter share no code. Truth is generated by fine-step Euler–Maruyama integration; the filter propagates a coarse closed-form model. The gap between them is a known quantity rather than an assumption: the simulator sits at a measured 2.98 per cent below the filter's model of the position variance at the default step count. A study that sampled the filter's own model to generate truth would have no gap at all and would be testing the filter against its assumptions.

Estimators see identical data. Random number streams are keyed to the scenario rather than to the estimator, so swapping EKF for CKF cannot perturb the measurement sequence. Any difference between them is the estimator.

Every result carries the smallest effect it could have detected. A verdict of "consistent" asserts only that no departure larger than the stated resolution was found, which is weaker than the absence of a departure, and several results in this study are inconclusive by that standard and are labelled so rather than quoted.

The consistency test is itself tested. It is run against a filter correct by construction and its false-alarm rate measured: 5.4 per cent against a 5 per cent design target. This matters because the test was wrong four separate times during development, and every one of those failures admitted filters that should have been rejected. One version passed a filter whose error exceeded its stated uncertainty by a factor of seven million.

4. What the comparison came out at

Two things had to be established in order: whether the filter is usable at all on this sensor, and then whether the choice of filter matters.

Table 2. Three close-range scenarios, 400 runs each. Range and bearing error are medians of the position error resolved along and across the line of sight. NEES is the departure of the estimation error from the uncertainty the filter claimed; zero means honest, positive means overconfident.

scenario	*range err*	*bearing err*	*EKF NEES*	*CKF NEES*	*track loss*
Intruder inspecting	155 m	6.3 m	+28137%	+13999%	40%
Intruder transiting	23 m	2.1 m	+239%	+12%	5%
Patrol flies straight	69 m	2.4 m	+383%	+49%	3%

The first column is the headline for the first question. Direction is estimated well and range is not, by a factor of twenty to thirty. That split is geometry, not tuning: a bearing constrains the across-track position directly and says nothing about distance, so range is only recoverable through the observer's own motion. Weaken that and it degrades: 23 metres against a target holding a course, 155 metres against one being flown by a person inspecting something, at a nominal 400 metres.

For the second question the picture depends on how hard the measurement nonlinearity is being pushed, which is set by the ratio of position uncertainty to range.

Table 3. Sweeping initial position uncertainty on a fixed geometry, 400 runs per point. Both filters see identical measurements.

p0 [m]	*EKF*	*CKF*	*error removed*	*track loss*
300	+3.4%	+3.1%	7%	0.0%
450	+8.2%	+5.9%	28%	0.0%
600	+19.2%	+8.7%	55%	0.0%
800	+43.0%	+8.6%	80%	2.0%
1000	+81.8%	+6.8%	92%	7.0%

At tight initialisation the two are indistinguishable and there is nothing to fix. As uncertainty widens the extended filter's linearisation degrades and the cubature filter does not, which is the expected behaviour and the reason the comparison was worth running at all.

Where the difference actually shows up

The gain is not evenly spread, and it is worth separating the two things a filter reports. Measured over 60 runs per point on surviving tracks:

Table 4. Position error and covariance honesty together. At tight initialisation the two filters are the same filter for practical purposes; the gap opens as the linearisation is pushed harder.

p0 [m]	*EKF error*	*CKF error*	*gap*	*EKF NEES*	*CKF NEES*
300	212 m	212 m	0.0%	4.10	4.09
600	333 m	326 m	2.1%	5.01	4.57
1000	424 m	370 m	12.7%	7.36	4.24

An earlier draft of this memo asserted that the cubature filter repairs the covariance and leaves the estimate alone. That is wrong, and the table is why it was checked rather than asserted. At wide initialisation the cubature filter is better on both counts: 13 per cent lower position error and a covariance that is honest rather than overconfident by a factor of nearly two.

What is true is that the covariance improves much more than the estimate does. At 600 metres the accuracy gain is 2 per cent while the overconfidence falls from 25 per cent to 14. So the case for switching is strongest where something downstream consumes the covariance, and weakest where only the point estimate is used and the initialisation is tight.

5. Things that did not work

Recorded because they consumed most of the time and because a study that reports only its successes is not worth much.

- *The obvious fix made things worse.* Holding the linearisation point fixed rather than following the estimate is the standard remedy for this class of problem. It returned an error of 4926 against a baseline of 4.55, because the method assumes stationary landmarks and this target moves. Iterating the update was also worse, and lost 34 per cent of tracks.
- *The first cubature rule gave frame-dependent answers.* The degree-3 rule recovered between 26 and 68 per cent of the same error under rotations of a physically unchanged problem. Its points sit on the coordinate axes, so it integrates no cross terms. The degree-5 rule returned a single value at every rotation.
- *Two sensor models were built and discarded.* A camera bolted to the airframe cannot hold a target with a narrow aperture; a gimbal makes a narrow aperture trivially usable and hides the route trade-off the simulator exists to show. The forward-fixed model was kept because a sentry on a fixed patrol carries one.
- *A three-actor intercept scenario was discarded.* A fast target against a slow sentry lost 69.5 per cent of its tracks, and the numbers first computed from it were survivor statistics presented as general.

6. What is not claimed

Nothing here is a new result. The weak observability of range from bearings is Lindgren and Gong, sharpened by Nardone and Aidala and by Fogel and Gavish. The degradation of the extended filter's linearisation with growing uncertainty is standard. The cubature rules are McNamee and Stenger's. The consistency statistics are Bar-Shalom's. The insensitivity of the innovation check to state-space error follows in two lines from the standard innovation covariance, and is quantified here rather than discovered.

What this project is, is a harness that lets those known effects be measured against each other quickly and checked afterwards. Six short notes accompany it, one per assumption the harness rests on, each stating the claim, citing the source, and naming the independent computation that verifies it. Writing them found five errors in my own descriptions, none in the code, which is the reason they exist.

7. Where it would go next

In order of value per unit of effort. Enabling the clutter and missed-detection paths, which are implemented and tested but not exercised in any reported result: about a day. Addressing track loss, which the cubature correction does not help and which is the largest remaining gap: about a week. Correcting the survivor conditioning that flatters every averaged figure here: about a day. Applying it to recorded flight data would transfer the harness rather than the specific numbers, and is the obvious next step for anyone wanting to know whether the simulation predicts the field.